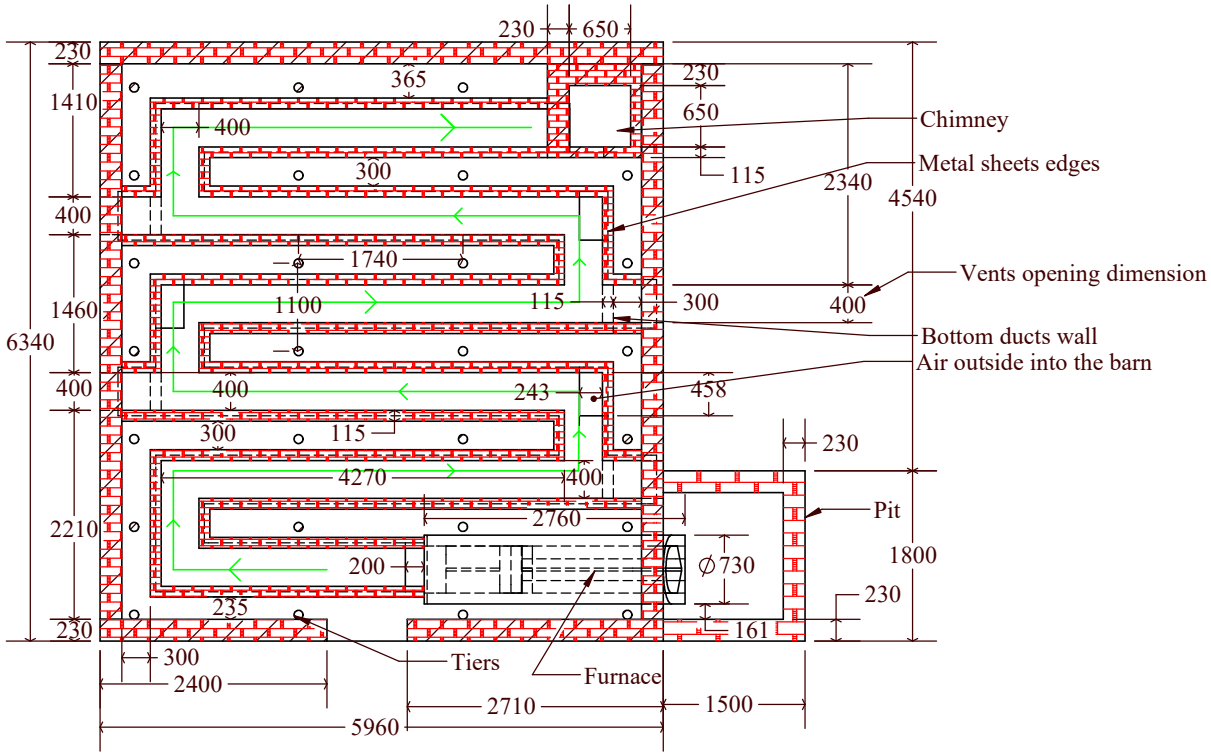
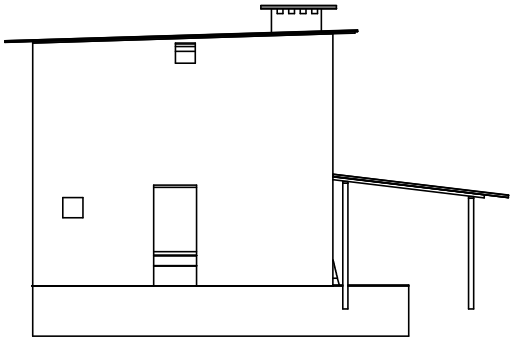
BILL OF QUANTITY (BOQ)				
#	DESCRIPTION	Qty	Unit price ($)	Total price ($)
1	Pc cement (50kg)	43 bags	11	473
2	Farm brick (230x115x78mm)	20 000	0.05	1000
3	Fire bricks (230x115x78mm)	25	9.43	235.75
4	Brick force 9" (roll 20m long)	18	1.36	24.48
5	Pit sand (50kg)	200	$7 per m^3	43.75
6	Anthill soil (50kg)	220	x	x
7	¾ stones (50kg)	12	$15 per m^3	5.63
8	Ducts Metal sheets (0.8mm)-2.4x1.2m	14	24	336
9	Vents and furnace door metal sheets (0.8mm) 2.4x1.2m	2	24	48
10	Timber (110mm diameter poles) 6m	35	9	315
11	U nails (40x40mm) 8 gauge	600	4.5 per 450	6
12	River sand (50kg)	8	$12 per m^3	3
13	250 micron Roofing and DPC plastic (230mmx 24m) & (7mx7m	1	100	100
14	Clips (short type talita)	900	1.4	1260
15	Thatching grass (roof insulation)	x	x	x
16	Hooks and Tying wire 8 gauge	1 rolls	78	78
17	105mm roofing poles 6m long	7	21	147
18	Roofing sheets (800mmx 6m) IBR	13	18	234
19	Barn door insulated (800mmx2m)	1	30	30
20	3" roofing nails (2kg)	1	6	6
	TOTAL			4345.61

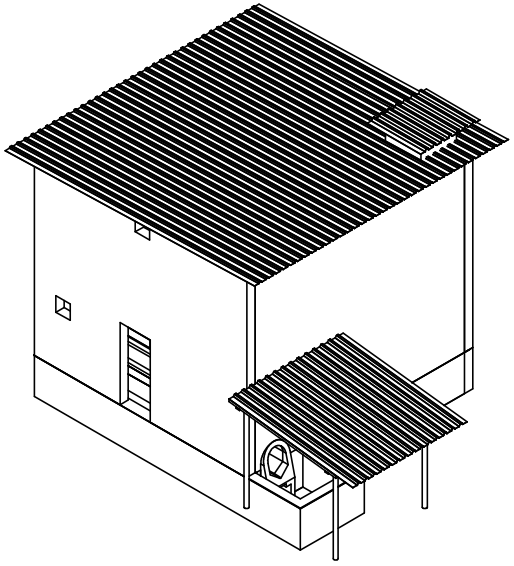
TITLE: KUTSAGA COUNTER CURRENT 1 (KCC1)	DESIGNED BY:R.SILINGWANI	DATE: 03/06/25	DRG.No:
	CHECKED BY: T.MAGUSHU	SCALE: 1:120	SHEET NO: 7
WEIGHT:	APPROVED BY:M.NGWENYA	QTY:	SHEET SIZE : A4



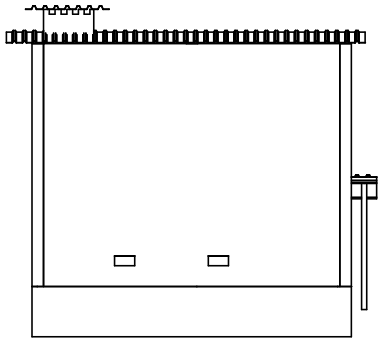
FLOOR PLAN
SCALE 1 : 80



FRONT ELEVATION

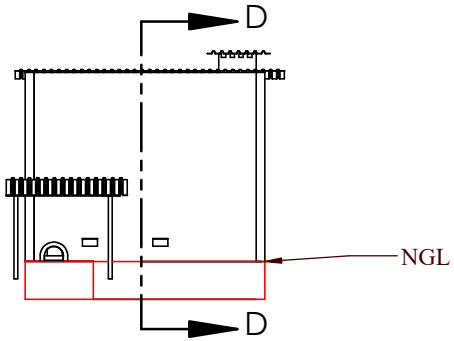


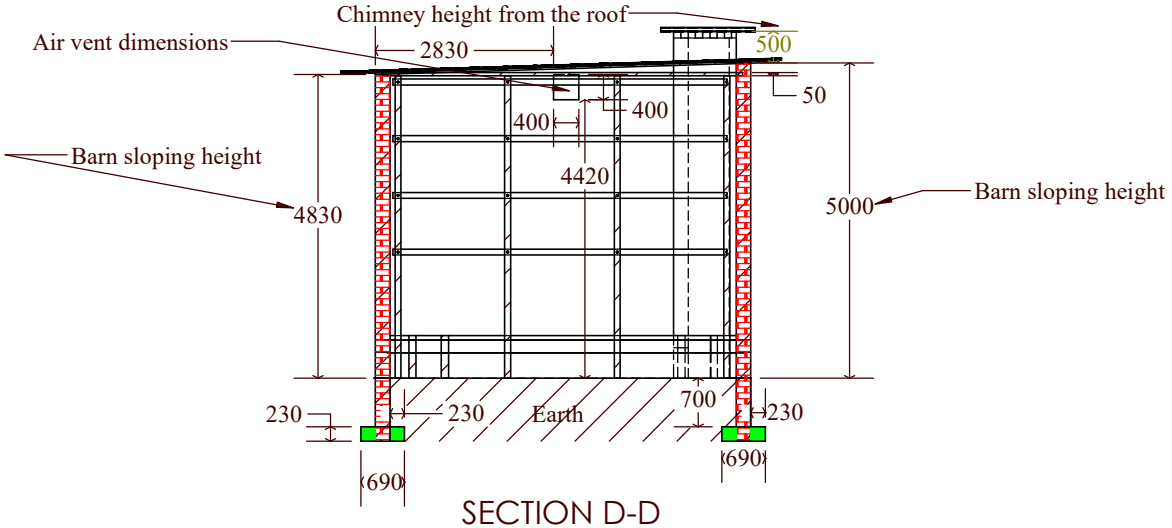
3D VIEW



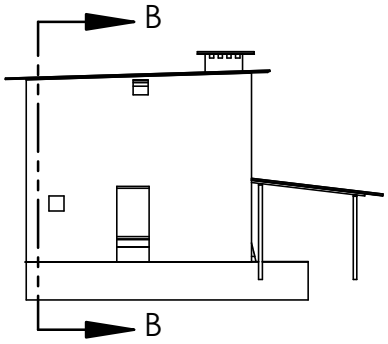
RIGHT ELEVATION

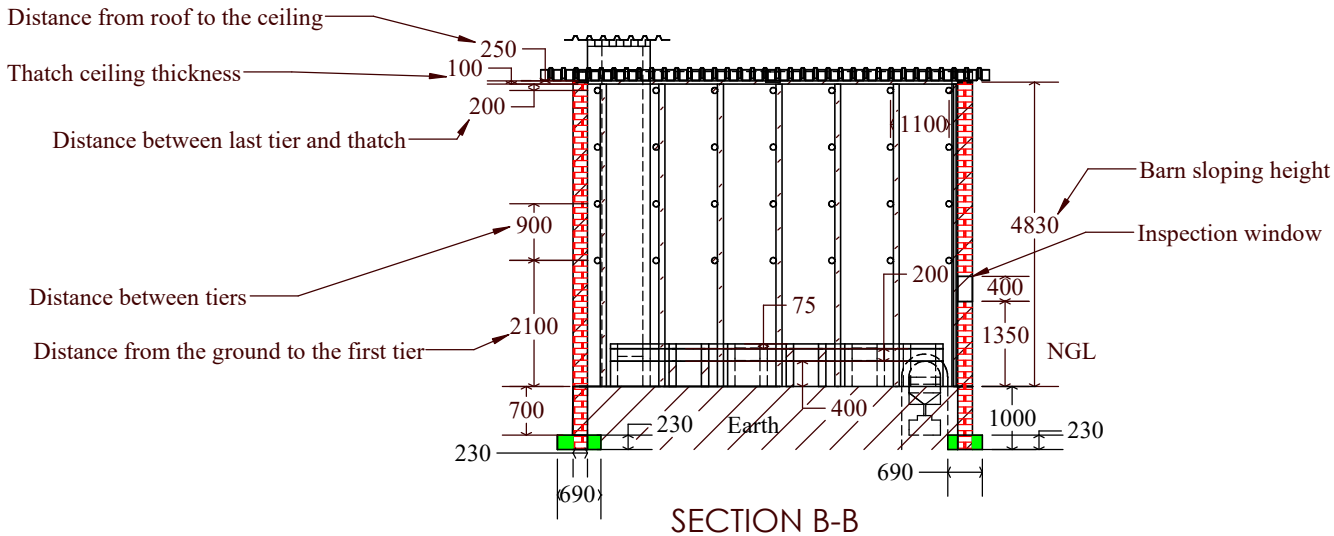
TITLE: KUTSAGA COUNTER CURRENT 1 KCC1	IMPROVED BY RONNIE SILINGWANI	DATE: 3 JUNE 2025	DRG.No:
	CHECKED BY: O. MUDADISI/ T.MAGUSHU	SCALE: 1:150	SHEET NO: 1
WEIGHT:	APPROVED BY: DR. MTHULISI NGWENYA	QTY:	SHEET SIZE : A4





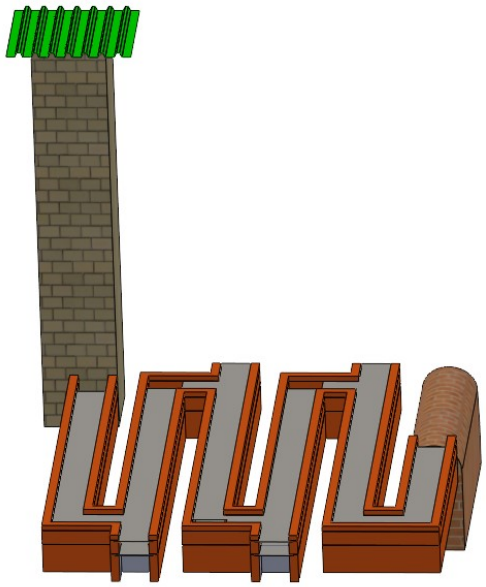
SECTION D-D
SCALE 1 : 120

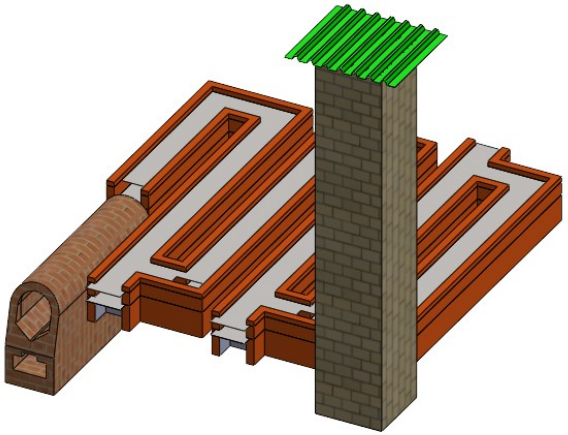


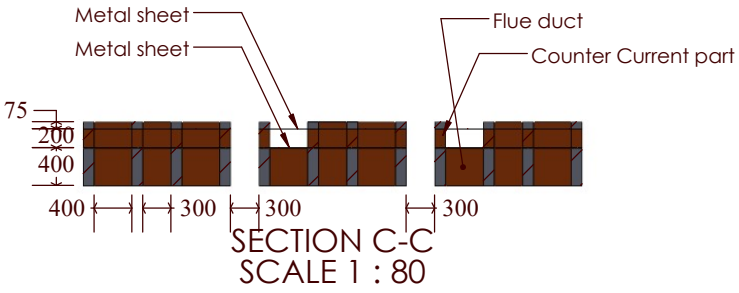


SECTION B-B
SCALE 1 : 120

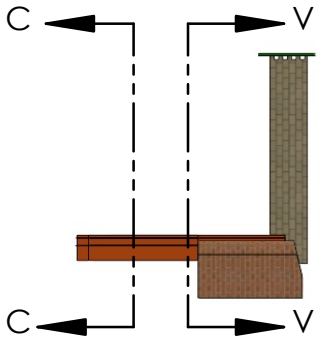
TITLE: KUTSAGA COUNTER CURRENT 1 KCC1	IMPROVED BY: RONNIE SILINGWANI	DATE: 3 JUNE 2025	DRG.No:
	CHECKED BY: O. MUDADISI/ T. MAGUSHU	SCALE: 1:200	SHEET NO: 2
WEIGHT:	APPROVED BY: DR. MTHULISI NGWENYA	QTY:	SHEET SIZE : A4

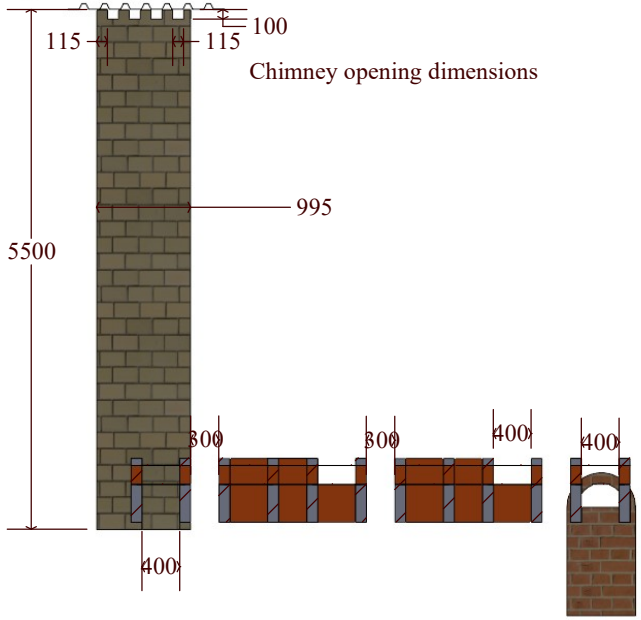






SECTION C-C
SCALE 1 : 80

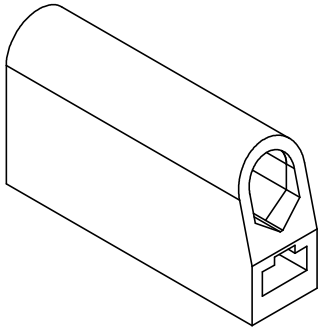




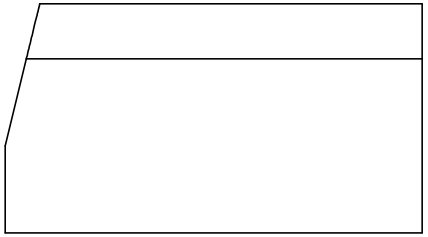
SECTION V-V
SCALE 1 : 80

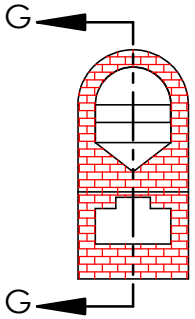
Heat exchanger

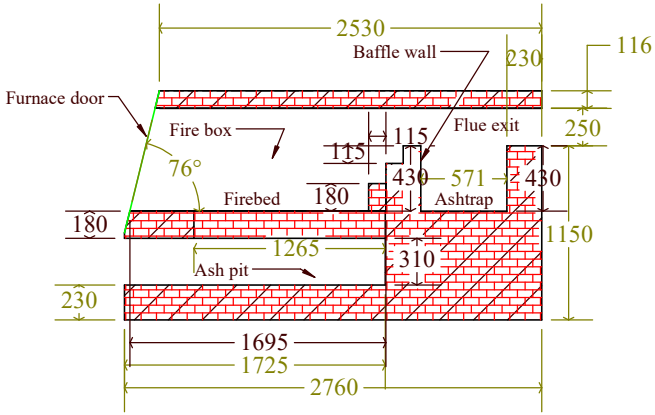
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	CHECKED BY: O. MUDADISI/ T. MAGUSHU	SCALE: 1:100	SHEET NO: 3
WEIGHT:	APPROVED BY: DR. MTHULISI NGWENYA	QTY:	SHEET SIZE : A4



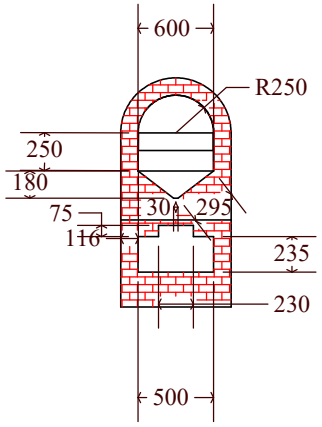
Furnace 3D view



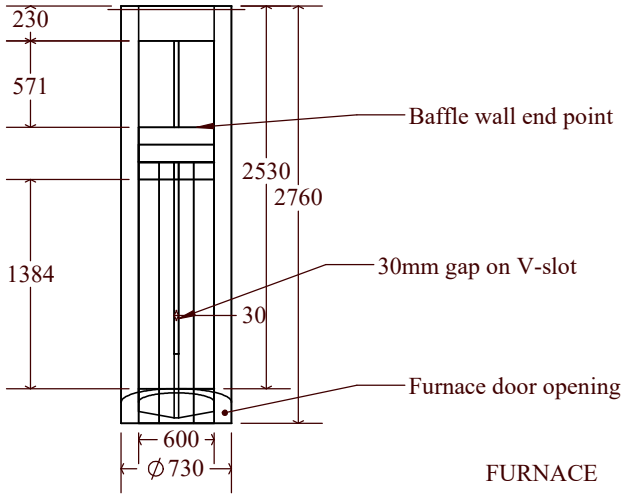




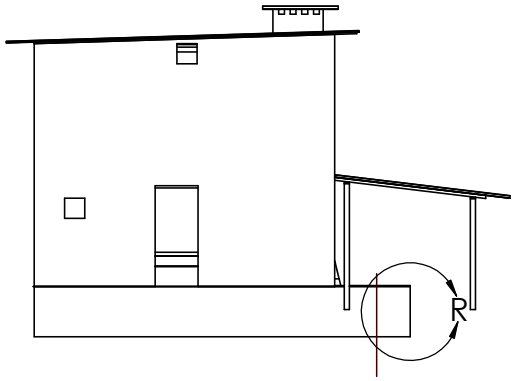
SECTION G-G
SCALE 1 : 50

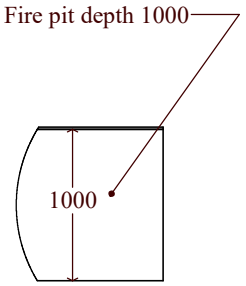


Furnace front view



Furnace internal view



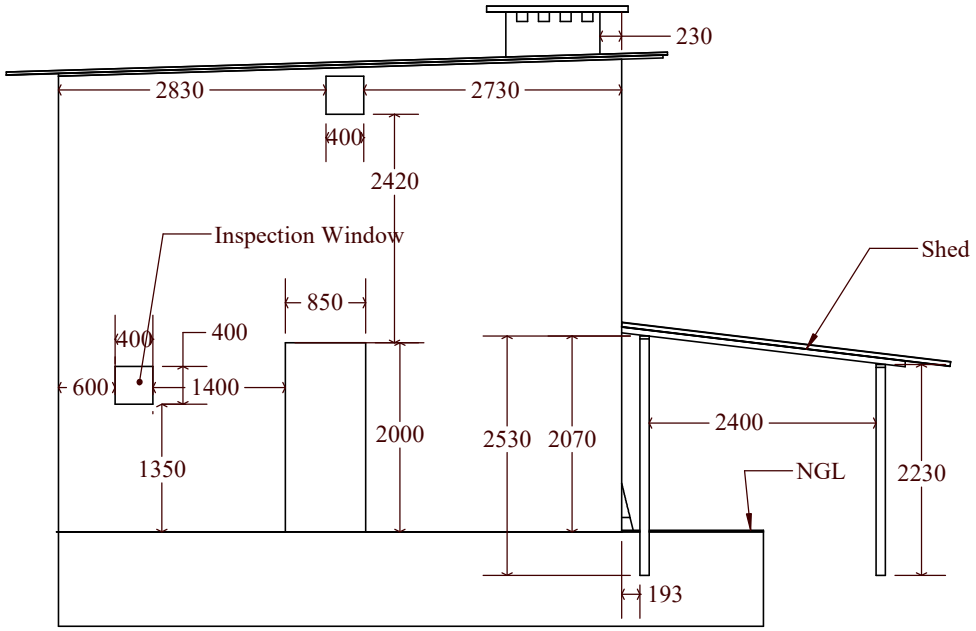


DETAIL R
SCALE 1 : 50

FURNACE

NB: ALL DIMENSIONS ARE IN MM

TITLE: KUTSAGA COUNTER CURRENT 1 KCC1	IMPROVED BY:RONNIE SILINGWANI	DATE: 3 JUNE 2025	DRG.No:
	CHECKED BY: O. MUDADISI/ T. MAGUSHU	SCALE: 1:50	SHEET NO: 4
WEIGHT:	APPROVED BY:DR. MTHULISI NGWENYA	QTY:	SHEET SIZE : A4



NOTES

Heat exchanger

- 0.6 mm galvanized sheet
- 1 mm galvanized sheet
- 700mm x 700mm furnace door (can be made from 1 mm metal

Roofing

- 105 mm roofing untreated poles
- 800mm x 3000mm roofing metal sheets (IBR)
- IBR ridges
- 800mm x 2000mm barn door insulated c/w frame

Ceiling

- Thatching grass(8mm thick)
- 250 micron black plastic - 7 m x 7 m
- Tying wire, 8 gauge

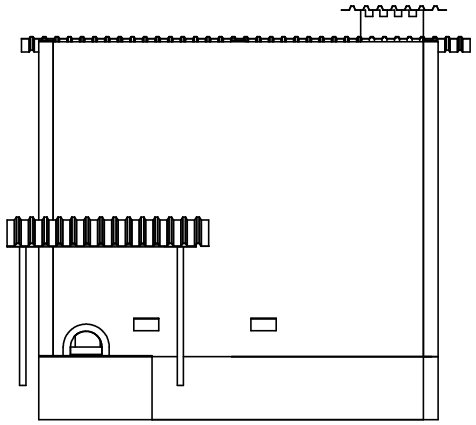
Timber

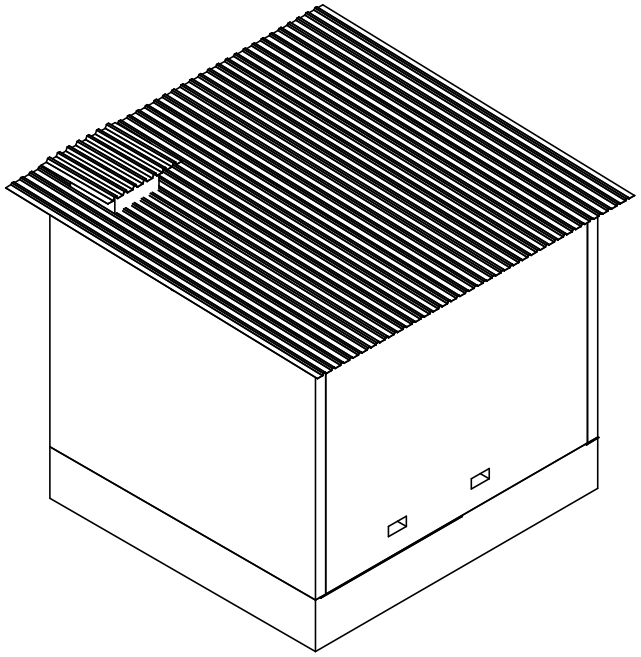
- 6.5 m long x 108mm - 120 mm pole diameter tiers

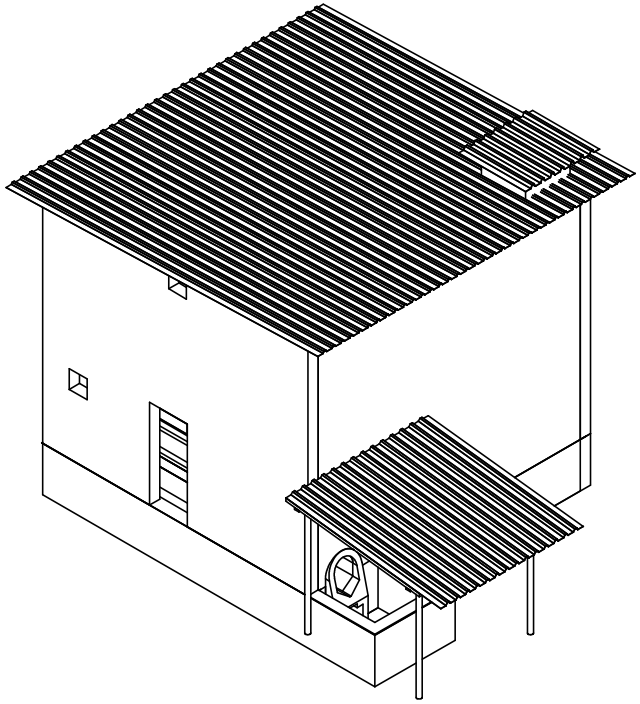
Farm bricks (230mm x 115mm x 75mm)

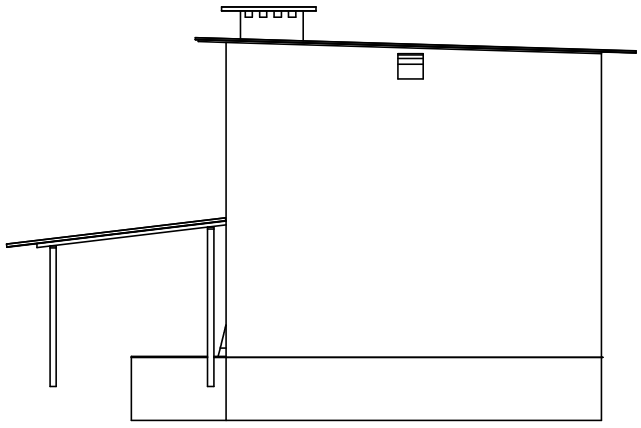
9'' brick force

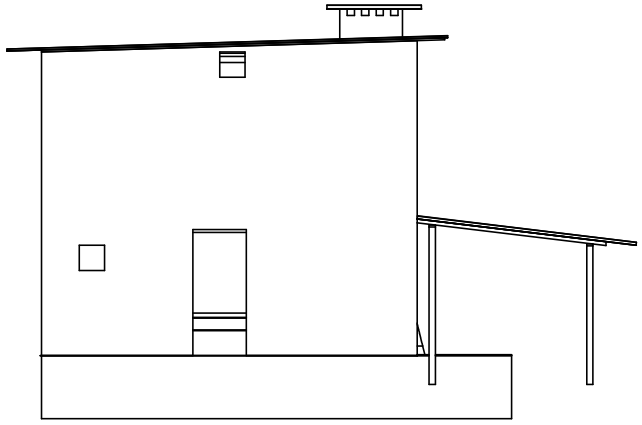
TITLE: KUTSAGA COUNTER CURRENT 1 KCC1	IMPROVED BY:RONNIE SILINGWANI	DATE: 3 JUNE 2025	DRG.No:
	CHECKED BY: O. MUDADISI/ T. MAGUSHU	SCALE:	SHEET NO: 5
WEIGHT:	APPROVED BY:DR. MTHULISI NGWENYA	QTY:	SHEET SIZE : A4

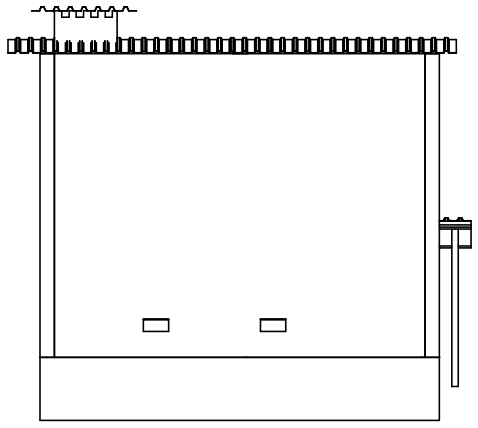












TITLE: KUTSAGA COUNTER CURRENT 1 KCC1	IMPROVED BY: RONNIE SILINGWANI	DATE:3 JUNE 2025	DRG.No:
	CHECKED BY: O. MUDADISI/ T. MAGUSHU	SCALE: 1:120	SHEET NO:6
WEIGHT:	APPROVED BY: DR. MTHULISI NGWENYA	QTY:	SHEET SIZE : A4